ADVENTURES IN BIG MATH · 5TH GRADE MATH

MISSION 03 OF 08 · WEEKS 1–4

Long Division

Two-digit divisors, and a remainder that means something.

THE BIG QUESTION

When you divide and there's something left over, when do you round up, round down, or keep the remainder?

HANDS-ON PROJECT

Road Trip Budget

Plan a real route: miles per day, gallons per tank, dollars per person. Every number divided by something.

FLUENCY GAME

Remainder Race

Roll a divisor, name the remainder before your opponent does. Wrong answers cost a turn.

AXIOM · MISSION COMMANDER, NINE TOURS

“Three buses for 100 people means three, not 3.33. The arithmetic gives you a number — the situation tells you what to do with it.”

Pre-Assessment

NAME ______________

DATE ______________

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Item F1 has no single right answer — it wants a decision and a reason.

STRAND A
DIVISION FACTS

A1 84 ÷ 12

A2 72 ÷ 8

STRAND B
ONE-DIGIT DIVISOR

B1 372 ÷ 3

B2 618 ÷ 6

STRAND C
TWO-DIGIT DIVISOR

C1 936 ÷ 24

C2 4,536 ÷ 21

STRAND D
ZEROS AND ESTIMATES

D1 7308 ÷ 36

D2 Estimate 3120 ÷ 39

STRAND E
REMAINDERS

E1 100 ÷ 7, with the remainder

E2 50 ÷ 8, with the remainder

STRAND F
INTERPRETING

F1 100 people, buses hold 30. How many buses?

F2 $50 split 4 ways, to the cent

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Division facts	A1 A2	Every Warm-Up tier	___ / 2	Warm-Up drops to 3 items all mission.
B · One-digit divisor	B1 B2	Week 1 · lesson 1.1	___ / 2	Day 1.1. B2 has an interior zero, so a pass there matters more than B1.
C · Two-digit divisor	C1 C2	Week 1 · lessons 1.2 to 1.4	___ / 2	Compress Week 1 to two days, but keep 1.3 — estimating the quotient is the skill everything else rests on.
D · Zeros and estimates	D1 D2	Week 2 · lessons 2.2 and 2.3	___ / 2	Day 2.2 only. A zero in the quotient is the single most common silent error in long division.
E · Remainders	E1 E2	Week 3 · lesson 3.1	___ / 2	Day 3.1. Finding a remainder is easy; deciding what it means is not.
F · Interpreting	F1 F2	Week 3 · lessons 3.2 to 3.4	___ / 2	Nothing. This is the Big Question — teach it even at 2/2.

ONE NOTE BEFORE YOU START

Watch F1 closely. A student who answers 3.33 has done the arithmetic correctly and missed the question — 100 people need four buses. That gap between a correct calculation and a correct answer is what Week 3 exists to close, and it is worth more than any amount of extra long-division drill.

NAME ___________

DATE ___________

One Digit, Cleanly

Divide, multiply, subtract, bring down. Say the four steps out loud.

Warm-Up

FACTS YOU ALREADY OWN

84 ÷ 12	72 ÷ 8	63 ÷ 9	120 ÷ 4	300 ÷ 5	360 ÷ 6
___	___	___	___	___	___

372 ÷ 3 · THE FOUR STEPS, TWICE

```
  124
 ____
| 3 ) 372
  - 3
  ----
     07
   - 6
  ----
     12
   - 12
  ----
      0
```

1 · How many 3s in 3? One. Write it above the 3.

2 · 1 × 3 = 3. Write it below.

3 · 3 − 3 = 0.

4 · Bring down the 7.

Then repeat: how many 3s in 7? Two, with 1 left. Bring down the 2 to make 12, and 12 ÷ 3 = 4 exactly.

Estimate first: 372 is close to 360, and 360 ÷ 3 = 120. The answer should be just over 120.

Core

ESTIMATE ABOVE EACH ONE FIRST

1. 372 ÷ 3

2. 618 ÷ 6

3. 896 ÷ 8

4. 1,236 ÷ 4

5. In item 2 the quotient has a zero in it. Which of the four steps produces that zero?

Challenge

NOT GRADED

C1. 6,036 ÷ 6. There are two zeros in the answer. Where do they come from?

C2. Multiply your answer to item 4 by 4. You should get 1,236 back. Why does that check work?

AXIOM: Multiply your answer back. Division is the only operation that hands you a free check, and it takes ten seconds.

Two-Digit Divisors

NAME ___________

DATE ___________

Round the divisor to guess, then check the guess and adjust.

Warm-Up

MULTIPLES OF THE DIVISOR

24 × 10	24 × 20	24 × 30	24 × 40	24 × 5	24 × 9
___	___	___	___	___	___

Write these six down before you start today's Core. A list of the divisor's multiples turns guessing into looking up.

936 ÷ 24 · GUESS, CHECK, ADJUST

```
  39
 24 ) 936
    - 72
      216
    - 216
        0
```

Round 24 to 20. How many 20s in 93? About four — but 24 × 4 = 96, too big. Try three: 24 × 3 = 72. That fits.

Bring down the 6 to make 216. How many 24s? Round again: 20 into 216 is about ten, but 24 × 10 = 240, too big. Try nine: 24 × 9 = 216 exactly.

The first guess being wrong is normal. Rounding the divisor down makes your guess too big; rounding it up makes it too small. Expect to adjust.

Core

1. 936 ÷ 24

2. 672 ÷ 21

3. 884 ÷ 34

4. 1,104 ÷ 46

5. For item 3 you rounded 34 to 30. Did that make your first guess too big or too small? Why?

Challenge

C1. 812 ÷ 29. Rounding 29 to 30 makes this quick — but which way will your guess be wrong?

C2. Which is easier to divide by, 25 or 29? Both are close to 30, so say what actually makes the difference.

NAME ___________

DATE ___________

Estimate the Quotient

Know roughly how big the answer is before you start dividing.

✦ Warm-Up

ROUND BOTH, THEN DIVIDE

400 ÷ 20	600 ÷ 30	900 ÷ 30	1200 ÷ 40	2000 ÷ 50	3600 ÷ 60
____	____	____	____	____	____

HOW MANY DIGITS WILL THE ANSWER HAVE?

4536 ÷ 21

21 × 100 = 2,100 — fits.
21 × 1000 = 21,000 — too big.
So the answer is in the hundreds.

812 ÷ 29

29 × 10 = 290 — fits.
29 × 100 = 2,900 — too big.
So the answer is in the tens.

Why it matters

If you expect a three-digit answer and get two, you dropped a digit somewhere. The estimate catches it before you finish.

◆ Core

ESTIMATE FIRST, THEN DIVIDE, THEN COMPARE

1. 4,536 ÷ 21

2. 3,120 ÷ 39

3. 8,464 ÷ 46

4. 2,451 ÷ 43

5. For each item above, write how many digits you expected in the answer. Did any of them surprise you?

★ Challenge

C1. Somebody divided 4,536 by 21 and got 26. How does the digit count tell you they went wrong without checking any arithmetic?

C2. 12,060 ÷ 60. Estimate the number of digits, then find the answer. The zero in the middle is not an accident.

NAME ___________

DATE ___________

Check by Multiplying

Quotient times divisor, plus the remainder, equals what you started with.

Warm-Up

MULTIPLY BACK

39×24	32×21	26×34	24×46	$14 \times 7 + 2$	$6 \times 8 + 2$
______	______	______	______	______	______

THE CHECK, WRITTEN OUT

$100 \div 7 = 14$ r 2

____

$14 \times 7 = 98$

$98 + 2 = 100$ ✓

Quotient × divisor + remainder = dividend. If that does not come out, one of the three numbers is wrong.

The remainder must always be **smaller than the divisor**. If you get 100 ÷ 7 = 13 r 9, the remainder is bigger than 7, which means 13 was too small a quotient. That rule alone catches most remainder mistakes.

Core

DIVIDE, THEN WRITE THE CHECK UNDERNEATH

1. $100 \div 7$

2. $500 \div 23$

3. $1{,}000 \div 31$

4. $750 \div 24$

5. Somebody says 500 ÷ 23 = 21 r 25. Without dividing, how do you know that is wrong?

Challenge

C1. A division has divisor 24 and remainder 7. What is the largest the remainder could have been?

C2. Find a number that leaves remainder 3 when divided by 8, and remainder 1 when divided by 5.

NAME ___________

DATE ___________

Remainder Race

Name the remainder before your opponent does.

How to play: one player names a number under 200. The other rolls a die for the divisor. Both race to say the remainder out loud — quotient not needed. First correct answer takes the round; a wrong answer costs you the next turn. If the remainder is zero, say "clean" instead.

Warm-Up · remainders only

17 ÷ 5	23 ÷ 4	50 ÷ 6	100 ÷ 3

Play eight rounds

ROUND	NUMBER	DIVISOR	REMAINDER	WON?
1				
2				
3				
4				
5				
6				
7				
8				

Challenge

C1. Dividing by 6, how many different remainders are possible? List them.

C2. Which divisor gave you the fastest answers, and why? It is not the smallest one.

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

QUIZ · end of week 2
TEST · end of week 4

Week 2 · Bigger Dividends QUIZ FRIDAY

Four-digit dividends, and the zero in the quotient that everybody drops.

MON 2.1
Four digits by two
Where the first digit of the quotient goes, and why.

TUE 2.2
Zeros in the quotient
The place that gets skipped is the place people lose.

WED 2.3
Estimate the quotient
A rough answer means a wrong one gets noticed.

THU 2.4
Divisors near a ten
Rounding 29 to 30 makes the guessing quick.

FRI
Mid-unit quiz
Page 10, then the Road Trip Budget begins.

Week 3 · What the Remainder Means

The mission's real content. Same division, four different right answers depending on what you asked.

MON 3.1
Find the remainder
It is always smaller than the divisor. Always.

TUE 3.2
Round up or down
Buses round up. Full boxes round down.

WED 3.3
Keep it as a fraction
Pizza does not round. Neither does money.

THU 3.4
Four questions, one division
$100 \div 7$ answered four ways, all correct.

FRI
Budget the miles
Miles per day, tanks per trip, dollars per person.

Week 4 · Proof UNIT TEST

The Road Trip Budget gets finished, and every rounding decision in it gets defended.

MON 4.1
Finish the budget
Every line item divided out and totalled.

TUE 4.2
Defend the rounding
Pick three lines and say why each rounded the way it did.

WED 4.3
Mixed review
Long division and remainder interpretation together.

THU
Error journal sweep
Fix only what repeats.

FRI
Mission 03 test
Pages 11–12. Twelve items plus one explanation.

NAME ______________

SCORE ______ / 8

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. 372 ÷ 3 ....

2. 936 ÷ 24 ....

3. 4,536 ÷ 21 ....

4. 7,308 ÷ 36 ....

5. Estimate 3,120 ÷ 39 ....

6. 812 ÷ 29 ....

7. 100 ÷ 7, with the remainder ....

8. Somebody says 500 ÷ 23 = 21 r 25. Explain how you know that is wrong without doing the division.

....

....

Marking note: item 8 wants the rule, not the right answer. Accept anything saying the remainder cannot be larger than the divisor.

Unit Test · Part 1

NAME ______________

DATE ______________

Items 1–8. No time limit. Check each answer by multiplying back.

1. 618 ÷ 6 ....

2. 884 ÷ 34 ....

3. 8,464 ÷ 46 ....

4. 9,072 ÷ 24 ....

5. 6,036 ÷ 6 ....

6. 1,178 ÷ 31 ....

7. 750 ÷ 24, with the remainder ....

8. Estimate 6,300 ÷ 68 to the nearest ten ....

CONTINUES ON PAGE 12

Unit Test · Part 2

Items 9–12 and the Big Question.

SCORE ______ / 12

BAND ______________

9. 100 people are going on a trip. Each bus holds 30. How many buses are needed?

10. 100 pencils go into boxes of 30. How many *full* boxes are there?

11. $100 is shared equally between 30 people. How much does each get, to the cent?

12. A 1,200-mile trip is driven over 5 days. If each day is the same distance, how far per day? If the driver wants no day over 200 miles, how many days does the trip take?

EXPLAIN YOUR THINKING · THE BIG QUESTION

When you divide and there's something left over, when do you round up, round down, or keep the remainder?

Items 9, 10 and 11 are the same division. Use them as your example. Scored on reasoning, not neatness.

11–12
Gold band. The Quotient Badge is earned.

9–10
Silver band. One reteach day, then the badge.

≤ 8
Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 · 7 | A2 · 9 | B1 · 124 | B2 · 103

C1 · 39 | C2 · 216 | D1 · 203 | D2 · 80

E1 · 14 r 2 | E2 · 6 r 2 | F1 · 4 buses | F2 · $12.50

F1 is the one that matters. An answer of 3 or 3.33 means the arithmetic worked and the question did not get read — mark it wrong and teach Week 3 in full.

Lesson 1.1 · One Digit, Cleanly

Warm-Up · 7 | 9 | 7 | 30 | 60 | 60

Core · 1 · 124 | 2 · 103 | 3 · 112 | 4 · 309

Item 5 · the divide step — 6 goes into 1 zero times, so a zero is written above before bringing the next digit down. C1 · 1,006; the zeros come from 6 not going into 0 and not going into 3. C2 · because multiplication undoes division, so the check recovers the dividend exactly when there is no remainder.

Lesson 1.2 · Two-Digit Divisors

Warm-Up · 240 | 480 | 720 | 960 | 120 | 216

Core · 1 · 39 | 2 · 32 | 3 · 26 | 4 · 24

Item 5 · too big, because dividing by a smaller number gives a larger quotient. C1 · 28; rounding 29 up to 30 makes the first guess too small. C2 · 25, because its multiples are familiar (25, 50, 75, 100) — closeness to a ten is not the only thing that matters.

TEACHER PAGE · ANSWER KEY

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 · Estimate the Quotient

Warm-Up · 20 | 20 | 30 | 30 | 40 | 60

Core · 1 · 216 | 2 · 80 | 3 · 184 | 4 · 57

Item 5 · three, two, three, two digits respectively. C1 · a two-digit answer where three were expected means a digit was dropped — 21 × 26 is only about 550, nowhere near 4,536. C2 · 201, and the zero appears because 60 does not go into 6 once the hundreds are taken.

Lesson 1.4 · Check by Multiplying

Warm-Up · 936 | 672 | 884 | 1104 | 100 | 50

Core · 1 · 14 r 2 | 2 · 21 r 17 | 3 · 32 r 8 | 4 · 31 r 6

Item 5 · the remainder 25 is larger than the divisor 23, so another whole 23 could still be taken out. C1 · 23, since the remainder must stay below the divisor. C2 · 11 works, and so does 51 — any number of the form 40n + 11.

Friday · Remainder Race

Warm-Up · 2 | 3 | 2 | 1

C1 · six remainders — 0, 1, 2, 3, 4, 5. Dividing by n always gives n possible remainders. C2 · usually 5 or 10, because the last digit tells you the answer immediately. Worth drawing out: 2 is fast for the same reason.

TEACHER PAGE · ANSWER KEY

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

1 · 124
2 · 39
3 · 216
4 · 203
5 · 80
6 · 28
7 · 14 r 2
8 · remainder rule

Unit test

1 · 103
2 · 26
3 · 184
4 · 378
5 · 1,006
6 · 38
7 · 31 r 6
8 · about 90
9 · 4 buses
10 · 3 full boxes

Items 11, 12 and the Big Question

11 · $3.33 each, with 10 cents left over — accept $3.33 with a note about the remainder. **12** · 240 miles a day over 5 days; capping at 200 miles means 6 days, since 1,200 ÷ 200 = 6 exactly. **Big Question** · full marks for an answer that ties the decision to the situation rather than the number: you round up when everything must be carried (buses), round down when only complete groups count (full boxes), and keep the remainder when the thing can be split (money, distance). Items 9, 10 and 11 are all 100 ÷ 30, which is the whole point — if a student notices that unprompted, say so.

Error journal · Mission 03

ONE LINE PER MISTAKE · STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read the whole list and fix only what appears twice.

DAY	WHAT I GOT WRONG, AND WHY